



Personalized musculoskeletal model based multi-muscle force analysis for amputees with transtibial prostheses

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ABSTRACT

Understanding the biomechanical effects of transtibial prostheses is essential for their optimization and for improving amputee mobility. This study investigated how different prostheses impact intra-limb muscle coordination and inter-limb muscle force symmetry using personalized musculoskeletal models based on the sDIMS platform. Five transtibial amputees participated in gait trials with passive prostheses and active prostheses under varying damping conditions. The results were compared to those of five able-bodied controls. Through multi-muscle force analysis, we quantified muscle force magnitudes, identified coordination patterns, and evaluated inter-limb symmetry of multiple lower limb muscles using correlation coefficients, cyclograms, and symmetry indices. The results revealed distinct compensatory strategies in the intact limb based on group-level analysis. Furthermore, personalized evaluation based on multi-muscle force analysis enabled effective assessment of different prosthesis configurations in a subject-specific manner. These findings suggest that the proposed framework can improve prosthetic parameter selection and support the development of personalized control strategies, offering practical guidance for transtibial prosthesis optimization.

1. Introduction

Transtibial amputations significantly alter gait biomechanics, leading to changes in joint kinematics, joint torques, muscle forces, and lower limb coordination. These changes often necessitate compensatory strategies in the intact limb (Silverman et al., 2008; Silverman and Neptune, 2012), which may increase the risk of long-term complications such as osteoarthritis (Lloyd et al., 2010) or secondary musculoskeletal injuries (Jayaraman et al., 2018). Despite advancements in prosthetic design, achieving a balanced and efficient gait remains a challenge for most amputees (Highsmith et al., 2016). These gait changes in amputees highlight the importance of optimizing prosthesis parameters to improve joint-level coordination and reduce long-term musculoskeletal risks. This raises critical questions about how prosthetic parameters influence biomechanical outcomes and how these influences can be optimized to address the individualized needs of amputees (Grabke et al., 2019), ensuring better movement coordination and long-term musculoskeletal health. Personalized musculoskeletal modeling provides a subject-specific simulation framework for estimating joint mechanics

and muscle forces, which can be used to evaluate the biomechanical impact of different prosthetic designs and parameter setting.

Current evaluations of lower limb prostheses primarily rely on sensor-derived experimental data (LeMoyne, 2016), including kinematic measurements, ground reaction forces (Viteckova et al., 2018), metabolic cost (Au et al., 2009), and surface electromyography (EMG). These sensor-derived approaches provide immediate, standardized measurements with established accuracy, offering valuable insights into gait performance. However, they are inherently limited in revealing musculoskeletal-level dynamics and inter-muscular coordination patterns. Musculoskeletal modeling has emerged as a powerful tool to address this gap through simulations, enabling non-invasive estimation of muscle forces and further biomechanical analysis. This modeling approach offers unique advantages, including the ability to compare relative force magnitudes across muscles, evaluate their contributions in human movement, and gain insights into deep muscles. These capabilities go beyond the scope of isolated sensor measurements. Nevertheless, biomechanical simulations are relatively complex and

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difficult to validate, and the calculation results of most simulations rely on sensor data. Therefore, the multimodal fusion between sensor measurements and model-based simulations represents the most promising solution and indicates the direction of future advancements. Sensor data provides the essential ground truth for model validation, while modeling extrapolates beyond measurable parameters to reveal underlying mechanisms. This integration ultimately provides a comprehensive understanding of dynamics and biomechanics during movement.

Several studies have highlighted the importance of muscle force and coordination in amputee gait. Russell Esposito and Miller (2018) found that maintaining muscle strength after amputation can help mitigate the increased metabolic cost of walking. Harandi et al. (2020) showed how intact limb muscles compensate for the loss of function in the residual limb. This adaptation can lead to altered muscle coordination in the non-amputated limb, potentially resulting in muscle atrophy (Xu et al., 2021). The function of prostheses in gait has also been discussed from a biomechanical perspective. Prostheses were observed to compensate for lost ankle function (Silverman and Neptune, 2012), particularly during sloped walking (Pickle et al., 2017). Furthermore, prosthetic design optimization has also become a focus. The influences of prosthetic foot stiffness (Fey et al., 2012) and prosthesis socket dynamics (LaPr   et al., 2018) have been considered. A spring-actuated biarticular prosthesis was proposed to replicate gastrocnemius function (Willson et al., 2021). Musculoskeletal modeling has played a crucial role in advancing these insights.

Despite the advancements discussed above, these studies often share common limitations. Most provide limited discussion and analysis of muscle force results, with the focus frequently restricted to a few individual muscles. To the best of our knowledge, current research has largely overlooked multi-muscle force coordination, particularly the interactions within and between the two lower limbs. However, understanding these relationships is essential for evaluating the biomechanical effects of prostheses on gait and optimizing their design (Zajac et al., 2003). Approaches such as correlation analysis and cyclograms (Goswami, 1998) provide valuable tools for quantifying and visualizing these relationships and symmetry, but their application in multi-muscle force analysis remains underexplored.

This study introduced a personalized musculoskeletal modeling framework for amputees to analyze the effects of different prosthesis types and parameter settings on gait biomechanics. More importantly, we applied multi-muscle force analysis including correlation analysis, symmetry indices, and cyclogram-based methods to assess muscle coordination patterns and inter-limb muscle force symmetry, providing deeper insights into compensatory strategies and prosthetic optimization.

2. Methods

2.1. Subjects and data collection

This study included five unilateral transtibial amputees (TTA) with an average age of 46.0 ± 8.8 years, height of 170.4 ± 1.4 cm, and body mass of 69.4 ± 4.3 kg. And the average residual limb ratio is $32.8\% \pm 3.9\%$. Five able-bodied participants (AB) also participated in the study for comparison. The able-bodied control group had an average age of 24.6 ± 1.4 years, height of 174.6 ± 2.6 cm, and body mass of 67.0 ± 8.0 kg. All the participants' characteristics are reported in Table 1. Although the two groups differed in age, they were similar in height and weight, which are more relevant for musculoskeletal modeling. Moreover, prior research (Gabell and Nayak, 1984) has shown that gait variability does not significantly differ between younger and older adults within the age range studied here. Four trials were performed for all amputee participants: one with their personalized passive prosthesis for daily use (P), and three with an active prosthesis (PKU-RoboTPro-I Pro) under varying damping settings: strong (active prosthesis with 80% of the

maximum damping rate, A80), medium (50%, A50), and weak (20%, A20). The extreme settings (0% and 100%) was avoided as it hindered natural gait performance of amputees. Details about this powered ankle robotic prosthesis can be found in our previous work (Wang et al., 2015; Feng et al., 2020). All participants, including the able-bodied controls, performed treadmill level walking trials at multiple speeds. To maintain a focused and interpretable analysis, 1.1 m/s was selected as a representative walking speed because it reflects a moderate-to-fast gait commonly used in biomechanical studies and yielded more stable gait patterns across participants. Results for other tested speeds are available in our previous work (Lu et al., 2025a).

The experimental data included kinematics, ground reaction forces (GRFs), and electromyography (EMG) signals. All experiments were conducted on a treadmill in a controlled laboratory environment, and participants were given time to adapt beforehand. The optical system and treadmill force plates were calibrated before each trial. EMG signal quality was ensured through standard skin preparation, including shaving and cleaning. Kinematic data were collected using a 12-camera optical motion capture system (MARS4H, NOKOV, China) with a sampling frequency of 100 Hz, GRF data were recorded using a treadmill embedded with a force plate (Gaitway-3D 150/50, H/P/COSMOS, Germany) and EMG signals were recorded using a wireless EMG system (Delsys Trigno Wireless EMG System, Delsys, USA) with 16 channels, both at a sampling frequency of 1000 Hz.

The kinematic data captured human movements through 28 reflective markers placed on the limbs, head, and torso, based on an improved version of the Helen Hayes hospital marker set (Murai, 2009). EMG electrodes were placed on key muscles of both limbs: on the intact leg, they covered the rectus femoris, vastus lateralis, biceps femoris caput longum, semitendinosus, tibialis anterior, gastrocnemius, and soleus according to Konrad (2005), while on the residual limb, only the thigh muscles were monitored.

2.2. Musculoskeletal model development

The musculoskeletal model employed in this study is a full-body model that covers the major musculotendon networks. This model is based on a customized software platform, sDIMS (Nakamura et al., 2005; Yamane et al., 2005), created by Yoshihiko Nakamura's research group at the University of Tokyo. Fig. 1 shows substantial information about the model, including the basic definition of model parameters, model input and output, calculation steps and methods. The computational pipeline for musculoskeletal modeling is also illustrated here, integrating the optimal motion capture system, force plate, and EMG sensors to estimate joint kinematics, dynamics, and muscle forces. Kinematic data from optical motion capture, GRFs from a force plate, and muscle activity data from EMG sensors constitute the model's inputs. The calculation steps begin with inverse kinematics, transforming marker positions to joint angles. Then Newton-Euler equations are used to calculate joint torques in inverse dynamics (Orin et al., 1979). Finally, optimization techniques based on muscle synergy and dynamic constraints are applied to address the muscle redundancy problem and estimate muscle forces (Ikegami et al., 2012). This integrated computational platform offers a robust framework for biomechanical analysis, combining multiple data sources into a unified workflow for human movement simulation. The calculated forces of representative muscles showed overall similarity and consistent trends with those reported in previous studies (Alexander et al., 2021).

For the transtibial amputees, subject-specific modifications were made to the musculoskeletal model to improve its accuracy and alignment with each individual's characteristics. The general musculoskeletal model was scaled according to each participant's anthropometric data. For the residual limbs, we adjusted the skeletal and musculotendon parameters according to the actual amputation site. Specifically, the mass properties and muscle parameters of the amputated limb were adjusted, and the shank portion was modified to include the residual limb, the socket, the pylon, and the main body of the prosthesis. Further details on the adjustments to model parameters and dynamics can be found in our previous work (Lu et al., 2025a).

Table 1
Basic information of experimental subjects.

Subject	Gender	Age	Weight	Height	Post-amputation	Amputation side	Residual limb ratio
TTA-1	Male	54	70	170	19	Left	40%
TTA-2	Male	32	73	171	11	Right	33%
TTA-3	Male	40	65	172	24	Left	30%
TTA-4	Male	55	64	168	33	Right	32%
TTA-5	Male	49	75	171	42	Right	29%
AB-1	Male	24	73	170	–	–	–
AB-2	Male	25	60	175	–	–	–
AB-3	Male	23	55	175	–	–	–
AB-4	Male	27	72	178	–	–	–
AB-5	Male	24	75	175	–	–	–

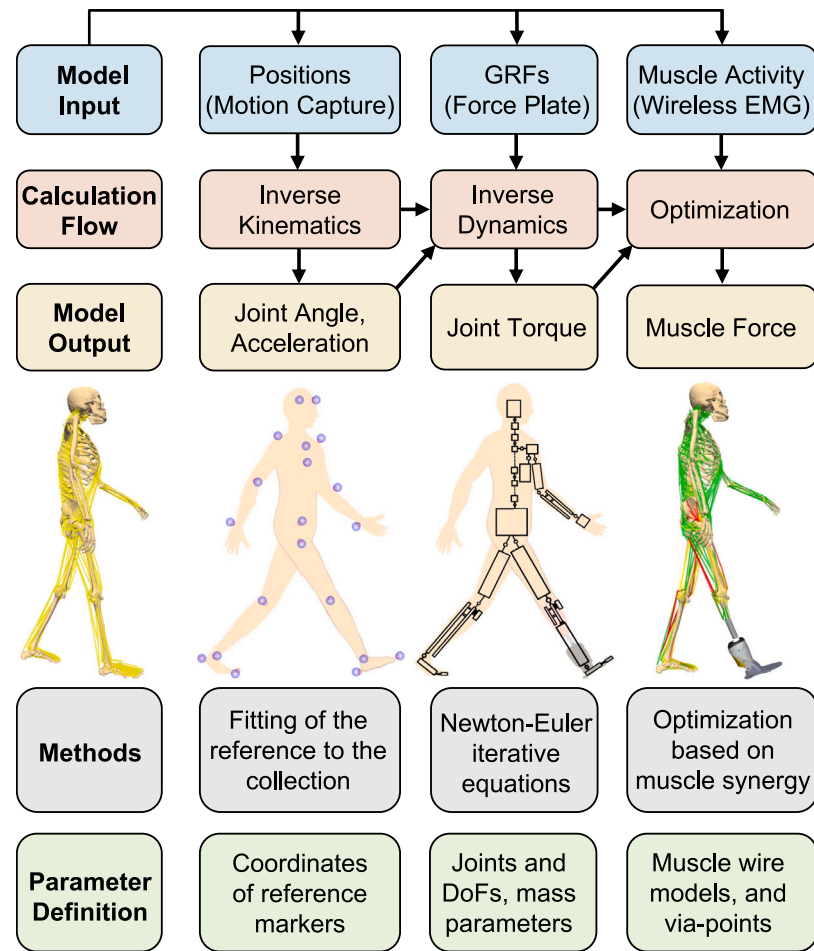


Fig. 1. Development of a full-body musculoskeletal simulation model for human movement in sDIMS.

2.3. Data processing and analysis

Optical, GRF, and EMG data were synchronized through a unified NI data acquisition system (National Instruments Corp., Austin, TX, USA), with additional alignment performed during post-processing to address differences in sampling frequency. Data were all processed using a customized MATLAB program (The MathWorks Inc., Natick, MA, USA). Raw EMG data were filtered using the 5th-order Butterworth band-pass filter with a cut-off of 20–450 Hz following motion-specific guidelines (Konrad, 2005). And the EMG signals were normalized to each subject's maximum voluntary contraction (MVC). Joint angles, torques, and muscle forces were estimated using the modified musculoskeletal model in sDIMS. The results for able-bodied participants were averaged across both limbs. Joint torques and muscle forces were normalized to individual body weight and interpolated over the 100% gait cycle. All values represent averages across multiple gait cycles.

The joint angles and torques of the hip, knee, and ankle in the sagittal plane were compared between the transtibial amputees and able-bodied participants. The muscle force outputs from the personalized model in the previous section were then systematically analyzed through multi-muscle force analysis methods (summary and comparison provided in Supplementary Table S2). Muscle forces were computed for the entire body. However, since our primary focus is on the lower limb muscles involved in walking, particularly those closely related to the prosthesis, the subsequent analysis focused on the muscles listed in Table 2. These muscles are crucial for sagittal plane movements of the hip, knee, and ankle, including hip flexion and extension, knee flexion and extension, and ankle dorsiflexion and plantarflexion. To analyze the magnitudes of muscle forces throughout the gait cycle, both peak and average values were extracted and compared across three groups: the intact and residual limbs of amputees, and healthy controls. Statistical significance was assessed by comparing (1) healthy controls

vs. amputees' intact limbs, (2) healthy controls vs. amputees' residual limbs, and (3) amputees' intact vs. residual limbs. Prior to testing, data normality was evaluated using the Shapiro–Wilk test. Depending on the distribution, either independent samples *t*-tests or Mann–Whitney *U* tests were conducted. *p*-values are reported as: * for $p < 0.05$, ** for $p < 0.01$, and *** for $p < 0.001$.

To provide a more comprehensive analysis of muscle force data, we also investigated the temporal characteristics and distribution of muscle forces across the entire gait cycle. While the preceding analyses were based on group-averaged data, this part focused on a single amputee and served as a personalized evaluation to compare the biomechanical effects of different prosthesis parameters. Specifically, we calculated the correlations of muscle forces within the intact limb to examine intra-limb coordination. To visualize inter-limb muscle force differences and relationships, we used cyclograms. Cyclograms are traditionally used to show joint coordination by plotting one joint's angle against another (Goswami, 1998). However, they also have broader utility in capturing dynamic relationships between paired signals over the gait cycle (Viteckova et al., 2018), including muscle forces (Kutilek et al., 2014). Based on muscle groupings and correlation patterns, cyclogram plots (Goswami, 1998) were generated to compare the residual and intact limbs under each prosthesis condition. In addition, we introduced a set of quantitative indicators to compare the effects of different prosthetic configurations. These included the peak and mean values of muscle forces across the gait cycle and the unitless Robinson symmetry index to assess inter-limb muscle force balance. Percentage changes in muscle forces were calculated by comparing each active prosthesis condition with the passive prosthesis.

3. Results

In this section, we first presented the joint angles, joint torques, and muscle forces, based on the averaged data from all amputee participants and healthy controls. These results serve as a basis for comparing the differences between the intact and residual limbs of transtibial amputees, as well as between amputees and healthy controls. Additionally, multi-muscle force analysis was conducted through correlations and cyclograms using one specific amputee subject (TTA-1) as an example. The single-subject results enable a highly personalized selection of prostheses or prosthetic parameters for specific needs.

3.1. Average group results

The basic kinematics and dynamics of the intact and residual limbs were compared through joint angles and joint torques, as shown in Fig. 2. H1, H2, K1, K2, A1, and A2 represent the time points at which peak joint torques occur for the hip, knee, and ankle joints, respectively. During the first 40% of the gait cycle, the curvature of the hip angle in the residual leg is larger than that in the able-bodied. Accordingly, the hip torque of the residual limb is greater during this phase of the gait cycle, with an increase observed at H1. Although knee joint angles show no significant differences between able-bodied individuals and both the intact and residual limbs of amputees, the knee torques in amputees exhibit a downward offset at K1 and K2, indicating underlying dynamic differences. In contrast, the ankle torques of both the intact and residual limbs are greater than those of able-bodied individuals. The large standard deviations of knee torques during the first 60% of the gait cycle and the large standard deviations of ankle torques from moment A1 to A2 indicate that there are substantial individual differences among amputees.

The muscle force results are categorized based on muscle grouping in Table 2, where the name abbreviations are also provided. The maximum and mean forces of muscles involved in hip flexion, hip extension, knee extension, knee flexion, ankle plantarflexion, and ankle dorsiflexion are shown in Fig. 3. The figure separates maximum (top half) and mean (bottom half) values along the *x*-axis, both plotted

Table 2

Muscle grouping of lower limbs based on joint motions.

Muscle groups	Included muscles
Hip flexors	Rectus Femoris (RF) Iliacus (ILI) Pectineus (PEC) Tensor Fasciae Latae (TFL) Sartorius (SAR)
Hip extensors	Biceps Femoris Caput Longum (BFL) Semitendinosus (ST) Gluteus Maximus (GMA) Adductor Magnus (AM) Semimembranosus (SM)
Knee extensors	Vastus Lateralis (VL) Vastus Medialis (VM) Rectus Femoris (RF)
Knee flexors	Biceps Femoris Caput Longum (BFL) Biceps Femoris Caput Breve (BFB) Semitendinosus (ST) Semimembranosus (SM)
Ankle plantarflexors	Tibialis Anterior (TA) Extensor Digitorum Longus (EDL) Extensor Hallucis Longus (EHL) Peroneus Tertius (PT)
Ankle dorsiflexors	Soleus (SOL) Gastrocnemius (GAS) Flexor Hallucis Longus (FHL) Flexor Digitorum Longus (FDL) Plantaris (PLA)

on a positive *y*-axis. Due to consistently lower forces of ankle plantarflexors and dorsiflexors on the residual limb, significant differences existed but were not marked in the figure for clarity. Overall, for most muscles, both the peak and mean muscle forces on the intact limb of amputees exceeded those of the corresponding muscles in able-bodied individuals, with the differences in mean values being more pronounced. Among hip flexors, RF and ILI contributed the most, while BFL dominated among hip extensors. Significant differences were found in both maximum and mean muscle forces for BFL, GMA, and AM between able-bodied participants and both limbs of amputees. For ILI, significant differences were observed between the intact and residual limbs of amputees. Hip extensor forces were generally higher on the residual limb, contrasting with hip flexors. The knee extensors did not exhibit significant differences between the residual and lower limbs. While among knee flexors, BFL contributed most, and BFB had significantly higher forces on the residual limb. For BFB, both maximum and mean forces differed significantly between able-bodied participants and the residual limb of amputees. The results of the ankle plantarflexors and dorsiflexors indicate that the muscles in the residual limb of transtibial amputees played a minimal role in walking. However, TA in the intact limb compensated notably, with the force indicators being significantly greater than those of able-bodied individuals. Among the ankle dorsiflexors, SOL remained the primary contributor during walking.

3.2. Individual case analysis

In this subsection, multi-muscle force analysis was performed using correlations and cyclograms for case study. TTA-1 was selected as a representative case to demonstrate the application of the proposed evaluation framework. Due to individual differences in amputation level, limb condition, and mobility, the results and parameter choices for TTA-1 are not intended to generalize across subjects. This case study serves to illustrate the feasibility of using multi-muscle force analysis for personalized prosthesis assessment. To assess inter-limb muscle coordination, the Pearson index was used to calculate the correlation of muscle forces during the gait cycle. This index is a discrete measure

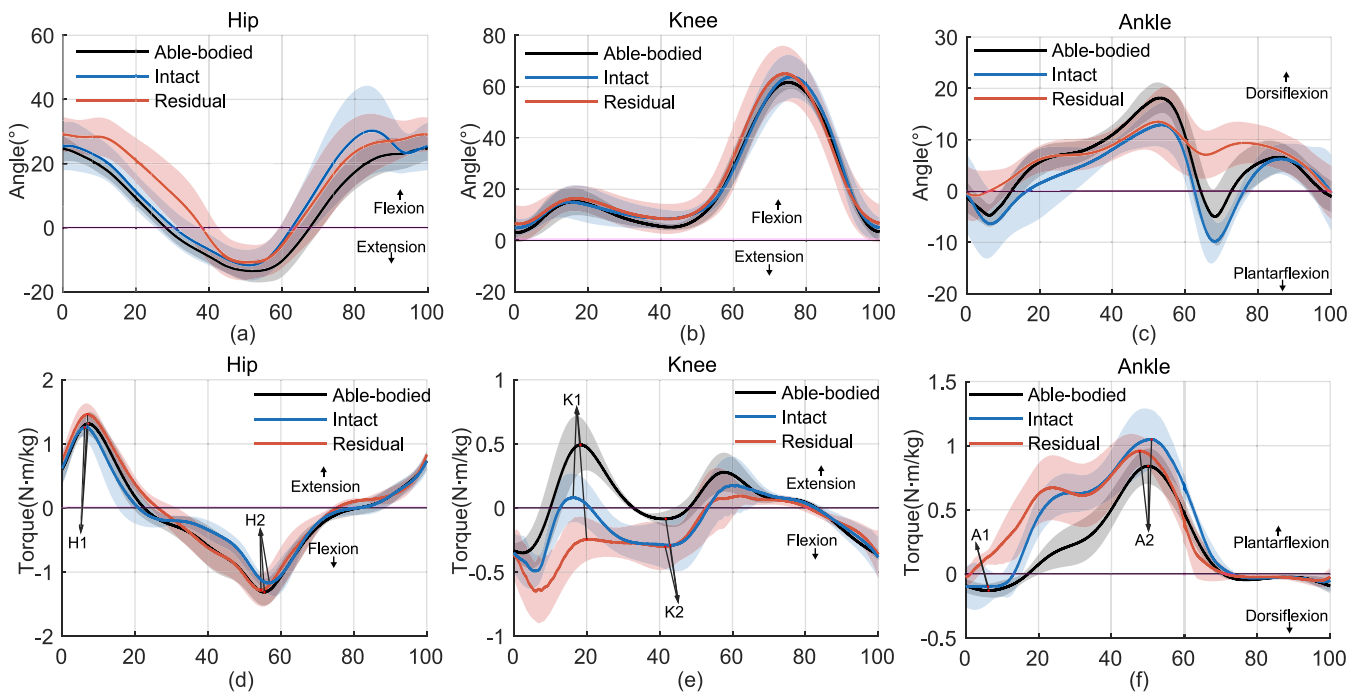


Fig. 2. Joint angles and joint torques in the sagittal plane at a walking speed of 1.1 m/s over 0%–100% gait cycle: (a) Hip angle, (b) Knee angle, (c) Ankle angle, (d) Hip torque, (e) Knee torque, (f) Ankle torque. H1, H2, K1, K2, A1, and A2 represent the time points of peak joint torques. The shaded areas indicate $\pm$ standard deviation from the mean. The amputee results presented here correspond to walking with their daily-use passive prostheses.

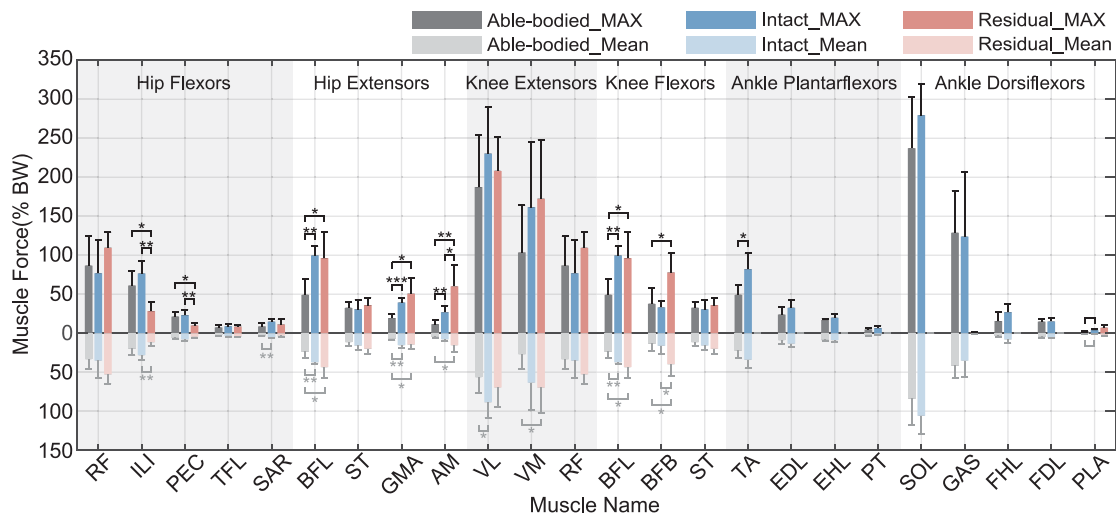


Fig. 3. The maximum and mean muscle forces of the primary muscles involved in lower limb joint motions during treadmill level walking at 1.1 m/s throughout the gait cycle. The amputee results presented here correspond to walking with their daily-use passive prostheses.

that quantifies the degree of positive or negative correlation between different muscle forces, providing a numerical representation of the balance in force generation between the two limbs. The correlation coefficients of muscle forces in the intact limb of TTA-1 while walking at 1.1 m/s with the daily-use passive prosthesis are presented in Fig. 4. The inter-muscle force correlations of hip flexors, hip extensors, knee extensors, knee flexors, ankle plantarflexors, and ankle dorsiflexors are separately framed with black solid lines. Most of the squares in these six frames were dark blue, which means the muscle forces within the muscle groups were positively correlated. The correlation between hip flexors and knee extensors, and the correlation between hip extensors and knee flexors are also separately framed, with black dashed lines. Most of the squares in the two frames were blue and darker, which

means hip flexors coordinated with knee extensors, and hip extensors coordinated with knee flexors during gait.

Based on these results, the summed muscle forces of six muscle groups were used for further analysis and comparison between the intact and residual limbs. Representative muscles were selected: RF for hip flexors, BFL and GMA for hip extensors, RF for knee extensors, BFL and BFB for knee flexors, TA and EHL for ankle plantarflexors, and SOL and FHL for ankle dorsiflexors. Cyclograms were generated to visualize muscle force relationships between the two limbs across the gait cycle, illustrating the symmetry in muscle force distribution. The proximity of the curves to the reference line (black line through the origin) indicates symmetry, with color intensity reflecting muscle force density.

Fig. 5 presents the summed and representative muscle forces for each group, with able-bodied data included as a reference. Hip flexors

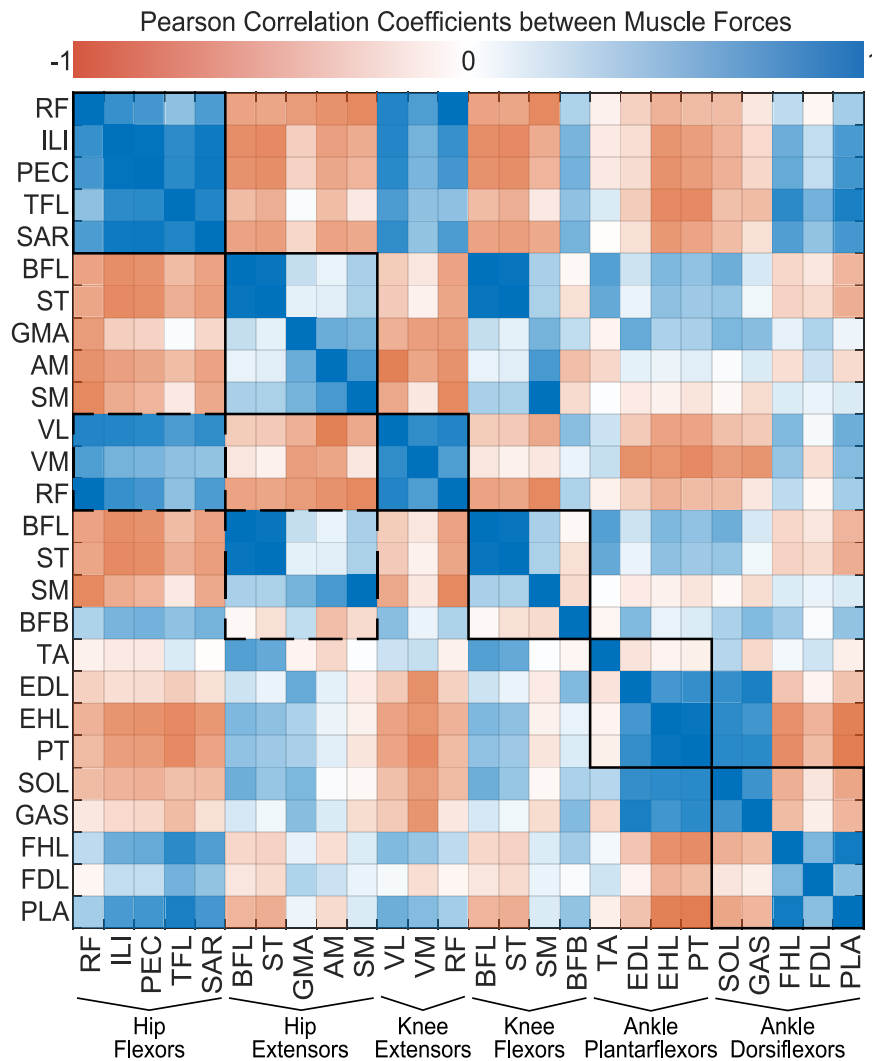


Fig. 4. Correlations of muscle forces in the intact lower limb (case-based analysis of participant TTA-1) while walking at 1.1 m/s with the daily-use passive prosthesis.

and knee extensors in the intact limb compensated significantly, as shown by the blue curves located mainly in the bottom-right quadrant in Fig. 5(a) and (c). Conversely, hip extensors and knee flexors in the residual limb contributed more during gait, as seen in Fig. 5(b) and (d). Maximum summed muscle forces for hip flexors and knee extensors were notably higher on the intact limb compared to able-bodied individuals. RF contributed primarily to hip flexors but less to knee extensors, while BFL played a major role in both hip extensors and knee flexors.

Specifically, the relationship of different curves can be used to identify magnitude differences and symmetry of muscle forces between the amputated and intact limbs during walking with different prosthetic devices. As shown in Fig. 6, each subfigure includes the summed group muscle forces while wearing passive prosthesis (P) and active prostheses with different damping rates: 20% (A20), 50% (A50), and 80% (A80) of the maximum damping capacity. Fig. 6(a) shows that the summed group muscle forces of hip flexors in both the residual and intact limbs are reduced when using an active prosthesis compared to a passive prosthesis. When wearing A80, the summed hip flexors muscle force in the residual limb was the lowest. As shown in Fig. 6(b) for hip extensors, the summed group muscle force in the residual limb became significantly bigger when wearing A20 and A50, and the muscle force varied much less when wearing A80 compared to wearing P. Fig. 6(c) shows that the summed group muscle forces of knee extensors had

better symmetry between the residual limb and the intact limb when wearing an active prosthesis. And when wearing A80, the peaks of summed knee extensor force in both limbs were the lowest. According to Fig. 6(d), when wearing an active prosthesis, the summed knee flexors force became bigger compared to wearing a passive prosthesis. Table 3 provides a quantitative summary of the muscle force patterns shown in Fig. 6, reporting peak and mean values as well as inter-limb symmetry across different prosthesis conditions. It serves to validate the visual findings from the cyclograms and facilitates direct comparisons between active and passive prostheses. Muscle force differences are shown as percentage changes relative to the passive prosthesis, with negative values indicating reductions. For the symmetry index, smaller values indicate better symmetry. As shown in Table 3, among the four prosthetic configurations, wearing A80 resulted in the largest number of muscle groups (7 out of 8) exhibiting reduced peak forces, with only the knee flexors on the intact limb showing no decrease. In cases where multiple active prostheses led to reductions, A80 generally produced the greatest decline. Regarding mean muscle forces, all eight muscle groups showed reductions under A50, while the residual limb muscles of all the four groups demonstrated greater decreases under A80. In terms of the muscle force symmetry index, wearing A80 improved symmetry in the hip extensors and knee extensors compared to the passive prosthesis, whereas both A20 and A50 led to reduced symmetry.

To evaluate the effects of different prostheses on the shank muscle forces, the maximum and mean forces of muscles involved in ankle

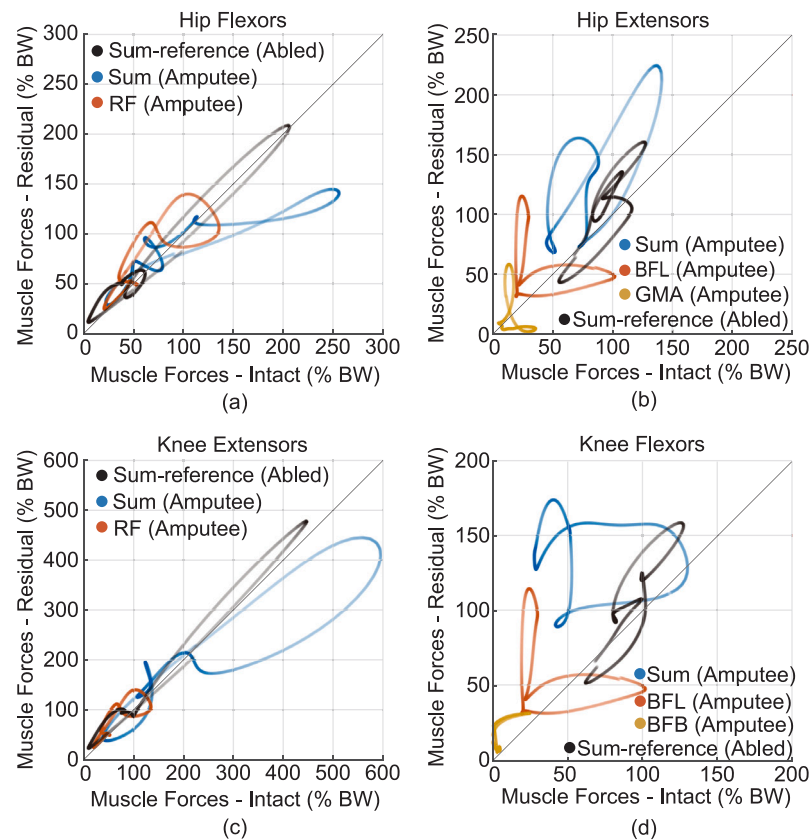


Fig. 5. Cyclograms of summed and representative muscle forces in different groups (case-based analysis of participant TTA-1) while walking at 1.1 m/s with the daily-use passive prosthesis. (a) Hip flexors, (b) Hip extensors, (c) Knee extensors, (d) Knee flexors. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3
Comparisons of group muscle forces for TTA-1 during treadmill level walking at 1.1 m/s under different prosthesis conditions.

Item	Muscle group	Side	A20 vs. P (%)	A50 vs. P (%)	A80 vs. P (%)
Peak	Hip flexors	Intact	-14.2	-11.3	-20.1
		Residual	-7.2	-19.8	-24.3
	Hip extensors	Intact	9.4	4.3	-14.1
		Residual	24.5	10.5	-7.4
	Knee extensors	Intact	-4.6	-8.6	-25.1
		Residual	9.9	6.7	-1.0
	Knee flexors	Intact	10.8	7.2	9.2
		Residual	6.8	5.5	-11.4
Average	Hip flexors	Intact	-23.9	-24.3	-6.3
		Residual	2.0	-14.4	-24.3
	Hip extensors	Intact	-6.2	-10.1	9.0
		Residual	6.1	-3.8	-17.9
	Knee extensors	Intact	-17.1	-17.4	-8.7
		Residual	13.6	-0.9	-14.9
	Knee flexors	Intact	-5.4	-14.3	7.5
		Residual	0.5	-5.8	-8.2
Item	Muscle group	P	A20	A50	A80
Symmetry	Hip flexors	0.403	0.759	0.781	0.766
	Hip Extensors	0.633	0.783	0.757	0.568
	Knee extensors	0.411	0.566	0.445	0.409
	Knee flexors	1.112	1.219	1.244	1.240

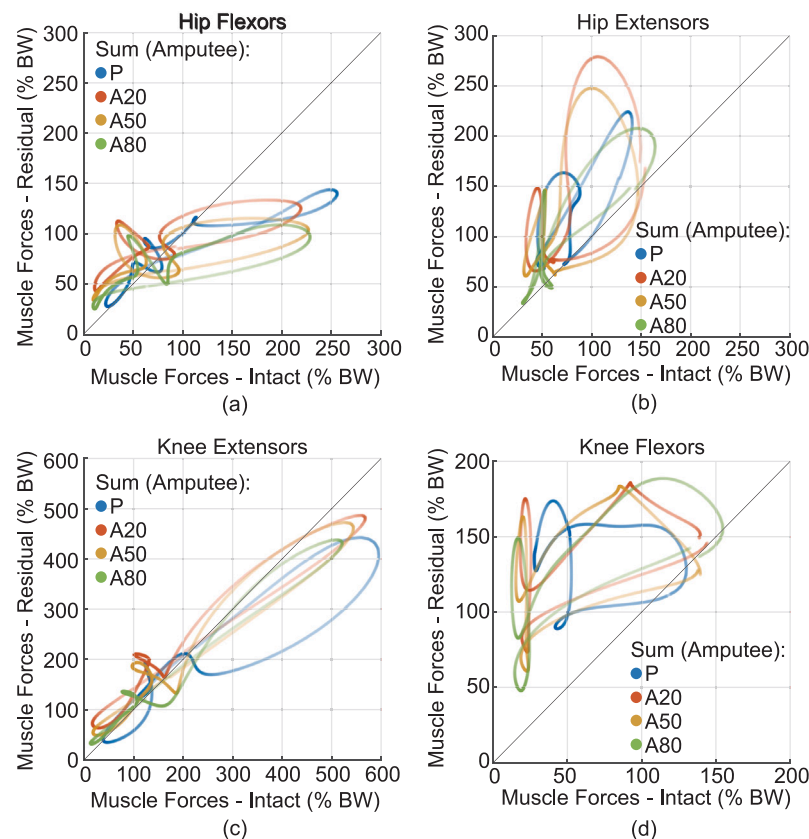


Fig. 6. Cyclograms of summed muscle forces with different prostheses (case-based analysis of participant TTA-1) while walking at 1.1 m/s. P: passive prosthesis; A80: active prosthesis with 80% of the maximum damping rate; same for A20 and A50. (a) Hip flexors, (b) Hip extensors, (c) Knee extensors, (d) Knee flexors.

plantarflexion and ankle dorsiflexion were shown in Fig. 7. As depicted in Fig. 7(a), both the summed force in ankle plantarflexors and the force of TA decreased while wearing active prostheses. And specifically, the summed force in ankle plantarflexors reached the minimum while wearing A80. In ankle dorsiflexors, both the summed force and the force of SOL increased while wearing active prostheses. However, while wearing A80, the increase of the summed force in ankle dorsiflexors and the increase of the mean force of SOL were minimal.

4. Discussion

In this study, an amputee-specific musculoskeletal model based on sDIMS was used for biomechanical computation and analysis. The joint angles, joint torques, and muscle forces of five transtibial amputees and five able-bodied participants were calculated. Comparisons of these results were made using subject-averaged data between the intact and residual limbs of the amputees, as well as healthy controls. Further analysis of muscle forces was based on the data of one specific amputee subject for personalized prosthesis selection. The correlation of muscle forces within and between muscle groups, the differences in the magnitude of muscle forces, and the symmetry of muscle forces between the intact and residual limbs were investigated.

Significant individual differences were observed between transtibial amputees on the joint angles and joint torques during walking in Fig. 2, which indicated the differences in kinematics and dynamics. These differences are largely due to the differences in amputation locations and prosthesis characteristics. During walking, the knee extensors and the ankle dorsiflexors made the most effort among the lower limb muscles according to Fig. 3. For transtibial amputees, the role of the shank muscles in the residual limb was limited. Correspondingly, the shank muscles in the intact limb had significant compensations. For thigh muscles, most muscle forces were either greater on the residual limb

or comparable between the two sides. A similar compensatory pattern was also observed by Voinescu et al. (2012), where the posterior thigh muscles of amputees generated higher forces compared to controls. While their analysis relied on individual line plots for each muscle, our study applied a multi-muscle approach based on both peak and mean values, enabling a more integrated assessment of inter-limb differences. Another previous work on amputee sloped walking reported hamstring-dominated compensations in the amputated limb without significant involvement of the GMA (Pickle et al., 2017). Whereas, our results during level-ground walking indicate that the GMA also played an important compensatory role. This discrepancy may be attributed to the different biomechanical demands of level versus sloped gait.

Synergistic muscles primarily function to coordinate specific movements and generally exhibit similar trends in muscle force patterns when executing a particular action (Safavynia et al., 2011). As indicated by the black dashed part in Fig. 4, the muscle forces of the hip flexors and knee extensors showed a strong inter-group correlation. A direct reason lies in the role of the RF, a typical biarticular muscle crossing both the hip and knee joints, contributing simultaneously to hip flexion and knee extension. This reflects motor control strategies that reduce energy expenditure and maintain dynamic stability during the swing phase (Barrett et al., 2007). Similarly, the hip extensors and knee flexors, particularly the biarticular hamstrings, also exhibited strong inter-group correlations, supporting joint coordination and balance during the stance phase. These findings align with prior research showing that amputees adapt their inter-joint coordination to maintain gait stability. For example, transfemoral amputees were reported to adopt altered hip-knee coordination in both limbs to compensate for reduced support on the amputated side and to ensure accurate foot placement (Xu et al., 2020). While this study analyzed joint-level kinematics, our analysis based on muscle forces provides complementary insights into compensatory control strategies at the muscular level. The

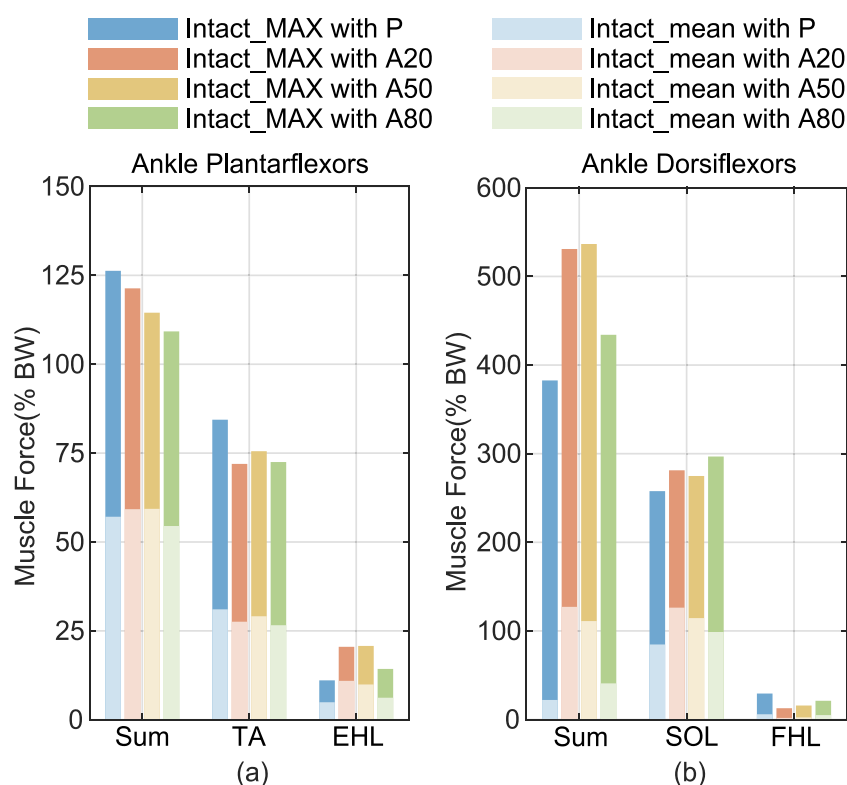


Fig. 7. Maximum and mean values of summed and representative muscle forces on the intact-side shank (case-based analysis of participant TTA-1) while walking at 1.1 m/s with different prostheses. TA and EHL are representative ankle plantarflexors; SOL and FHL are representative ankle dorsiflexors. P: passive prosthesis; A80: active prosthesis with 80% of the maximum damping rate; same for A20 and A50. (a) Ankle plantarflexors, (b) Ankle dorsiflexors.

intra-group correlations observed for knee flexors, ankle plantarflexors, and dorsiflexors further demonstrate how synergy patterns shift across gait phases in response to biomechanical demands.

The cyclogram provides an intuitive and compact way to visualize bilateral muscle force magnitude and symmetry throughout the gait cycle. Compared to line plots, which require multiple curves to separately display each limb and prosthesis condition, cyclograms can integrate the information of two limbs into a single two-dimensional trajectory, allowing for more efficient comparison across muscles and conditions. Although temporal details are not explicitly represented, cyclograms preserve the overall distribution characteristics of muscle forces and effectively reveal inter-limb coordination patterns. In this study, cyclograms were used to compare thigh muscle force distributions between residual and intact limbs under different prosthesis settings. According to the results in Fig. 6, Table 3, and Fig. 7, A80 appeared to be the most suitable configuration for subject TTA-1 when walking at 1.1 m/s. The passive prosthesis used by TTA-1 has no movement degrees of freedom in ankle joint, which limits its ability to adapt to dynamic gait conditions and impairs push-off efficiency. However, since the subject was accustomed to this device, its performance was relatively stable. In contrast, the 20% and 50% damping settings in the active prosthesis allowed excessive ankle motion, which may have led to reduced stability and increased variability in muscle activation. The 80% damping condition appears to provide an effective balance by moderating energy dissipation while maintaining sufficient joint support and responsiveness during late stance. Notably, Table 3 shows that under the A80 condition, the average muscle forces of all four thigh muscle groups in the residual limb were significantly reduced. This reduction is likely attributed to the enhanced push-off control provided by the active prosthesis, which reduces the compensatory effort required from residual limb muscles during the late stance and early swing phases. Consistent with previous findings that prostheses were large contributors to knee contact forces via dynamic coupling (Silverman and Neptune, 2014),

our results indicate that active prosthesis configurations can effectively redistribute joint loading and contribute to muscular unloading in the residual limb.

Several limitations should be considered when interpreting our results. First, the relatively small sample size, age difference between two subject groups, and the single-gender composition of participants may affect the generalizability of the findings. Although prior research suggests that age has limited impact on gait variability within this range (Gabell and Nayak, 1984), other studies have identified age-related differences in certain kinematic and kinetic parameters (Nigg et al., 1994), albeit with small absolute effects. Therefore, the age difference remains a potential source of bias and should be considered when interpreting the results. Besides, the individual case analysis focused on one specific transtibial amputee walking at 1.1 m/s, which may not fully represent general walking strategies. Moreover, the only adjusted parameter in the active prosthesis was damping. Future work will investigate the effects of adjusting parameters such as torque, stiffness, and control strategies to identify the optimal settings. Additionally, evaluation results based on traditional methods will be compared with those based on muscle force analysis. To advance the proposed framework, future research should integrate real-time muscle force estimation (Durandau et al., 2017) with closed-loop feedback control, supported by machine learning and data-driven methods (De Vree and Carloni, 2021). Recent works by Xu et al. (2022) and Lu et al. (2025b) have demonstrated how machine learning can significantly enhance biomechanical simulations across various scenarios. Building on these advances, EMG-driven or sensor-based estimators trained on experimentally collected data and musculoskeletal model outputs could be developed to rapidly infer muscle forces during walking. Such estimators could provide near-instantaneous inputs to update the prosthesis model in real time, allowing adaptive control strategies that adjust damping or other parameters dynamically based on the user's current gait. As highlighted by Pizzolato et al. (2019), the personalized

digital twins integrating real-time NMS modeling and machine learning offers a particular promise for creating clinically viable solutions. This combined approach would also support individualized prosthesis tuning for daily use.

In this study, we utilized personalized musculoskeletal models to evaluate the effects of transtibial prostheses on amputee gait. By customizing the models to individual amputations and incorporating multi-muscle analyses, we explored muscle coordination and muscle force symmetry between the residual and intact limbs. Through correlation analyses and cyclograms, we gained a deeper understanding of both inter-limb and intra-limb biomechanics of amputees. The evaluation of different damping configurations demonstrated that the A80 active prosthesis yielded the most favorable results during fast walking in the case study. However, this study does not aim to recommend a specific damping setting for all users. Given the variability among amputee individuals, prosthesis parameters should be selected based on personalized evaluation. The proposed framework offers a foundation for subject-specific analysis and may support clinical decision-making in future prosthesis tuning and optimization.

CRedit authorship contribution statement

Yuwen Lu: Writing – original draft, Methodology, Investigation, Formal analysis. **Yan Huang:** Writing – review & editing, Validation, Project administration, Methodology. **Wei Jin:** Visualization, Validation, Formal analysis. **Qining Wang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jbiomech.2025.112913>.

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